Progress update

Making quantum circuits shorter, and proving how short they can get

Exact ground truth for local rewriting, and a general hardware-aware optimiser

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Building on B. Rosenhahn, T. J. Osborne, C. Hirche, *Optimization driven quantum circuit reduction*, New J. Phys. **27**, 104509 (2025)

The story in one slide

The starting point

- A recent paper shortens circuits by swapping small pieces for shorter ones from a precomputed table.
- It is compared only with Qiskit, on random circuits.

What we found

- Every gate it uses is **Clifford**, so the true optimum is computable.
- Against it, the method is **3–4.5**$\times$ too long, and we can say why.

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What we built

- A **general optimiser** for real native gates, connectivity and calibrated noise, exact including phase.

Where it stands

- Proven optimum on **35/40** of the paper's 3-qubit circuits.
- After Qiskit L3 on 58 algorithms it never adds a 2q gate and removes **13 %** of gates (Heron).
- Up to 95 qubits it removes **19 %** on top of Qiskit L3, every output certified.

Part I

The method we start from

Replace short windows of a circuit with shorter equivalents from a precomputed database.

Why shorten circuits at all?

- On today's hardware every gate costs fidelity. Roughly,

$$\Pr[\text{success}] \approx \prod_{\text{gates } g} \big(1 - e(g)\big),$$

  the **ESP**. Two-qubit errors are 10–100× single-qubit ones.
- Compiling to a device's **native gates** leaves redundancy.
- Goal: an *equivalent* circuit with fewer, or cheaper, native gates.

Figure from arXiv:2502.14715, Fig. 1

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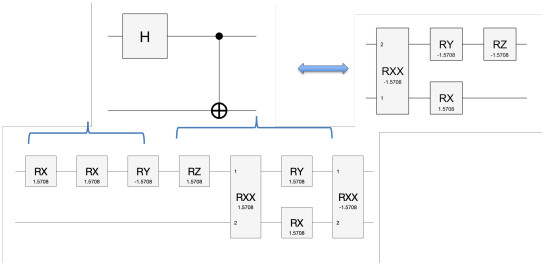
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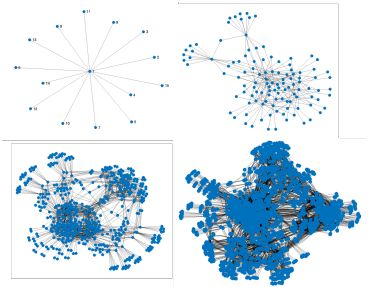
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H + CNOT compiled to ion-trap gates (8 gates) and the shortest equivalent (4 gates).

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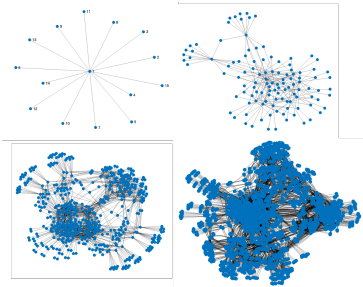
The method: a database and a sliding window



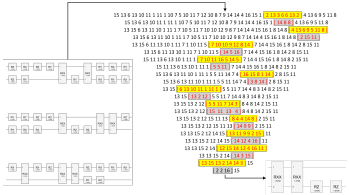
Compute graph. Nodes are circuits (as matrices), edges add one gate. Everything within radius r of the identity is stored with its shortest form. The paper uses $r = 5$–6.

Figures from arXiv:2502.14715, Figs. 2 and 3

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Compute graph. Nodes are circuits (as matrices), edges add one gate. Everything within radius r of the identity is stored with its shortest form. The paper uses $r = 5$–6.



Window. Take 2–8 gates on ≤ 3 wires, look them up, substitute a shorter word, shuffle commuting gates, repeat.

Takeaway. Two knobs decide everything later: the window length w and the database radius r.

Figures from arXiv:2502.14715, Figs. 2 and 3

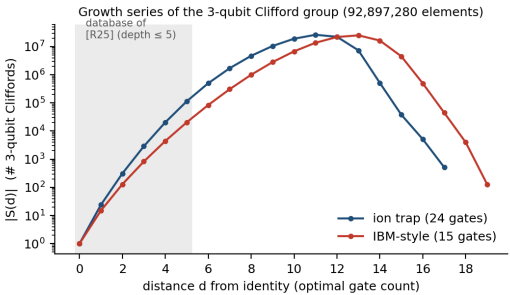
Part II

Exact ground truth

Every gate the paper uses is Clifford, so the search space is a finite group we can enumerate completely.

Every gate in the paper is Clifford

- $\pm\frac{\pi}{2}$ rotations, $R_{XX}(\pm\frac{\pi}{2})$ and CZ map Paulis to Paulis.
- So every circuit is an element of the finite **Clifford group**: 24 elements for 1 qubit, 11,520 for 2, **92,897,280** for 3.
- We enumerated all of them for 3 qubits and stored the exact shortest length of each (25 min, 6 GB).

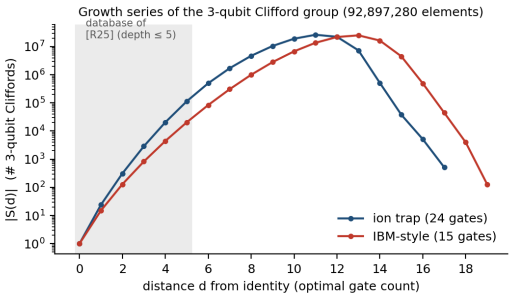


Elements by optimal length (3 qubits). The paper's depth-5 database (shaded) covers 0.15 % of the group.

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Takeaway. Optimality is no longer vague. Each test circuit has a proven shortest length.



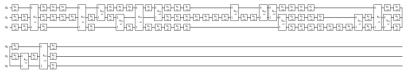
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How far from optimal is the paper's method?

3 qubits, 300 random gates, 20 circuits per row

	optimum	paper's method	Qiskit L3
ion trap	**10.4**	47.1	127.2
IBM-style	**12.2**	36.8	104.0

(a) input: random 3-qubit word over the ion alphabet, 80 gates



(b) original method (database radius 5): 24 gates



(c) proven optimum (exact Cayley table): 11 gates



A typical circuit: 80 gates in, the paper's method stops at 24, the proven optimum is 11.

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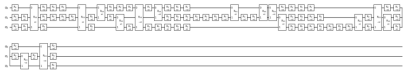
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The gap

The method's output is **4.5**× (ion) and **3.0**× (IBM-style) the optimum, on its own benchmark. Our port matches its published number (47.1 vs. 45), so this is the method, not our code.

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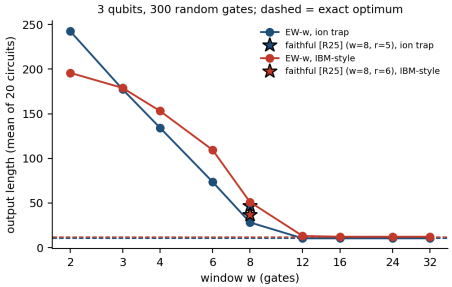


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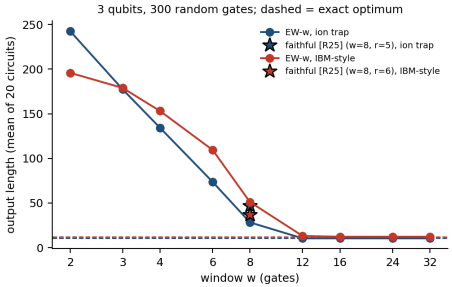
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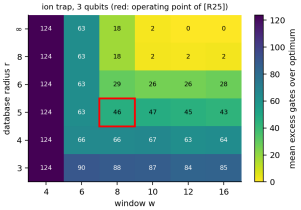


With exact lookups, a window of $w \geq 12$ (ion) or 16 (IBM-style) reaches the proven optimum on all 40 circuits.

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Mean excess over the optimum by window w and radius r.

The paper's $(8, 5)$ in red.

- Larger window at $r = 5$: excess 46 $\rightarrow$ 43.
- Larger database at $w = 8$: 46 $\rightarrow$ 18.
- $w \geq 12$ and $r \geq 8$: excess ≤ 2.

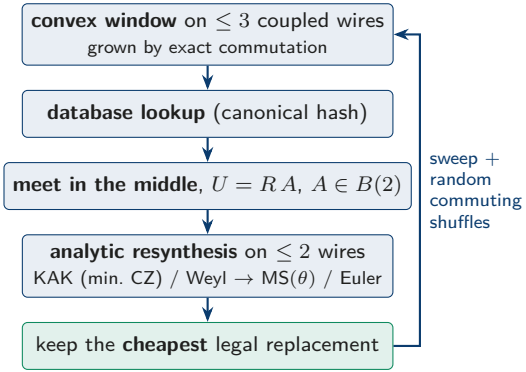
Takeaway. The database radius is the binding constraint. Our theorem predicts that both knobs must be large enough.

Part III

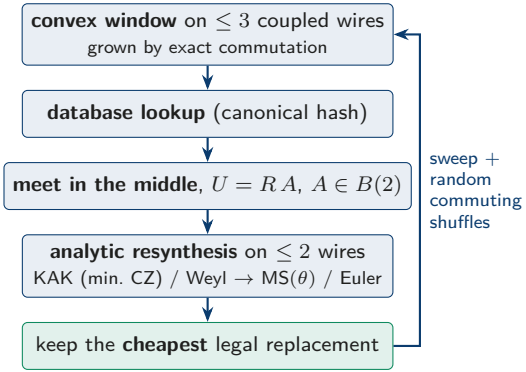
A general optimiser

Drop the Clifford crutch and add continuous angles, connectivity, calibrated noise, and exactness including phase.

The engine


convex window on ≤ 3 coupled wires
grown by exact commutation
database lookup (canonical hash)
meet in the middle, $U = R\,A$, $A \in B(2)$
analytic resynthesis on ≤ 2 wires
KAK (min. CZ) / Weyl $\rightarrow$ MS(θ) / Euler
keep the **cheapest** legal replacement
sweep + random commuting shuffles

The engine



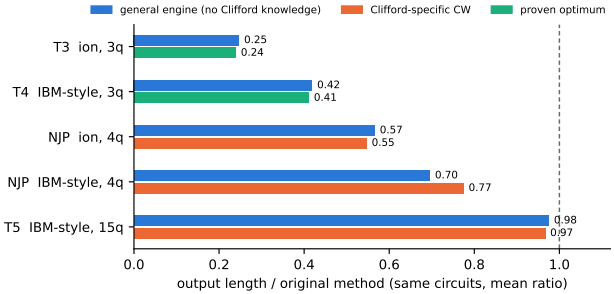
- **Cost** $c_{\mathrm{cal}}(g) = -\log(1 - e(g))$ from calibration, so minimising cost maximises ESP.
- **Exact including global phase**: phases are tracked and repaired once at the end.
- **Never worse** than its input, by construction.
- **Any gate set**: databases of 0.3–31 M classes for IBM Heron and trapped ions.
- Runs *after* any compiler, as a post-pass.

Part IV

Results

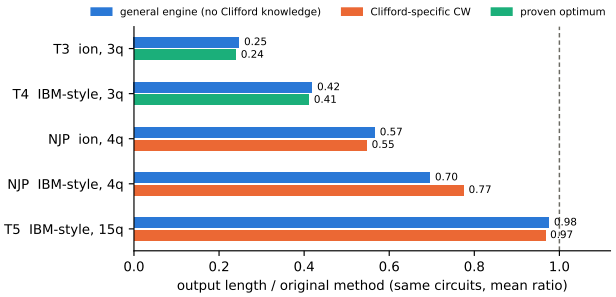
The paper's benchmarks against ground truth, 60 algorithm circuits on hardware models, and circuits up to 100 qubits.

Result 1. The paper's benchmarks, without Clifford knowledge



Output length relative to the paper's method, same circuits and seeds.

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Output length relative to the paper's method, same circuits and seeds.

- **3 qubits.** Proven optimum on **35/40**, mean 10.8 vs. optimum 10.4.
- **4 qubits.** On par with the Clifford-specific optimiser.
- **15 qubits.** All methods within 4 % (random circuits leave little local structure).
- **Clifford+T, 2 qubits.** Optimal on 99/99 random and 41/41 planted circuits.

Takeaway. Knowing the group is not what matters. Large databases and convex windows are.

Result 2. 60 algorithm circuits, relative to Qiskit level 3

	IBM Heron (58 circuits)			**trapped ion** (58 circuits)	
pipeline	2q gates	all gates	ESP	2q gates	all gates
TKET	×1.02	×1.71	×1.16	×1.27	×1.79
ours, exact, from Qiskit L1	×1.00	**×0.84**	**×1.21**	**×0.95**	**×0.76**
ours, exact, from Qiskit L3	**×0.98**	**×0.87**	×1.03	×1.00	×1.00
BQSKit L3 on L1 (approximate)	**×0.85**	×1.20	**×1.22**	**×0.94**	**×0.71**
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Nam/VOQC, QASMBench, QFT, QAOA, Trotter, EfficientSU2, Steane, random. 2–12 qubits. Ratios of summed counts, geometric mean for ESP.

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- **Stacked on L3** ours never adds a 2q gate and never lowers ESP.
- **BQSKit** finds better 2q structure but is approximate and leaves many 1q gates. Ours on top keeps its 2q savings and removes **16 %** more gates on Heron, exactly.

Nam/VOQC, QASMBench, QFT, QAOA, Trotter, EfficientSU2, Steane, random. 2–12 qubits. Ratios of summed counts, geometric mean for ESP.

Which objective? Let noisy simulation decide

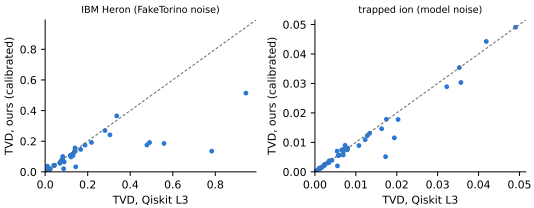
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Calibrated beats Qiskit L3 on **39/49** Heron circuits.

Takeaway. Counting 2q gates with weight 10 is almost as good as full calibration, a useful proxy when no calibration exists.

Result 3. Large circuits, 14–100 qubits

Relative to Qiskit L3

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	2q	all	ESP*	2q	all
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ours from L1	×1.05	**×0.78**	×0.43	**×0.96**	**×0.68**
ours from L3	**×0.98**	**×0.81**	**×1.12**	×1.00	×1.00

45 Nam/VOQC circuits, 88 of 90 runs done. pf circuits excluded (Qiskit L3 drops their 10^{-7} rad rotations). *Over the 13 Heron circuits whose ESP is at least 10^{-3}.

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45 Nam/VOQC circuits, 88 of 90 runs done. pf circuits excluded (Qiskit L3 drops their 10^{-7} rad rotations). *Over the 13 Heron circuits whose ESP is at least 10^{-3}.

- **Stacked on L3, Heron.** 19 % fewer gates, fewer CZ on 20/22 circuits, never more.
- **From L1.** Fewer gates than L3 everywhere, but lower ESP: at this scale Qiskit's L3 layout matters more than what we remove.
- **Cost.** Minutes per circuit, up to 6 h for the largest, single-threaded Python.
- **Running.** BQSKit on this suite (up to 40 qubits).

Takeaway. Larger circuits leave more to remove, and the guarantees hold unchanged.

Every output is verified, even beyond 25 qubits

- **Up to 24 qubits.** $\|(U - V)\psi\|$ on random states: 250/250 paper outputs, 720/720 suite outputs, 57 large outputs (max $1.4 \cdot 10^{-10}$).
- **Clifford, Clifford+T.** Exact arithmetic.
- **Beyond.** Statevector is out of reach, and MQT QCEC gave no answer on routed 23-qubit adders.

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Replay certificates

- The optimiser logs every step. An independent checker replays it with small matrices on a few wires.
- **171/176** large outputs certified, up to 544,922 steps, at most 6.8 s each.
- 5 more were rejected by an older checker limit and are being re-checked.

Takeaway. Checking a whole circuit becomes many tiny local checks, so cost grows with the steps, not the width.

Part V

Wrap-up

Where it stands and what comes next.

Limitations

- Noise models, no real-hardware runs yet.
- Windows of at most 3 qubits, single-threaded Python, hours at 100 qubits.
- Baselines are Qiskit, TKET and BQSKit at default settings and one seed.

Limitations and next steps

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Next

- BQSKit on the large suite and the 5 re-checks (running).
- VOQC, Quartz and QUESO on the Nam gate set.
- Several seeds with error bars, and an ablation of window, radius, convexity and cost.
- Real IBM hardware, last.

Summary

1. The original method is **3–4.5**$\times$ off the proven optimum, and the database radius is why.
2. A general optimiser with deep databases and convex windows reaches the optimum on **35/40** circuits without knowing the group.
3. On real algorithms it is a safe post-pass: after Qiskit L3 it never adds a 2q gate and removes 13–19 % of gates, and after BQSKit it removes 16 % more.
4. Every output is verified, by simulation or by replay certificate.